

The background features a dark field with a pattern of fine red dots. Overlaid on this is a series of wavy, horizontal red lines that create a sense of motion or a signal waveform. The text is centered and rendered in a bold, white, sans-serif font.

Spread spectrum techniques in disposable drones for anti drone evasion



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Drone and Robot Builder
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Hacker and hacking communities
co-organizer

Coding & Cybersecurity
instructor & mentor



Context: a World in War



Drones as War Elements



Drones as War Elements

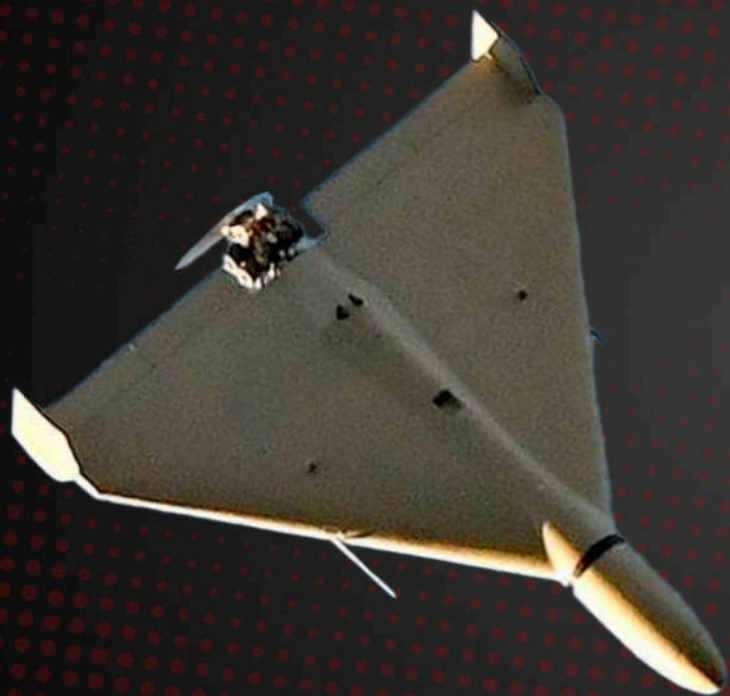
- Directed weapons
- Unmanned
- Elements for symmetric warfare
- Means of strategy and tactics for real-time covert operations

Drones as War Elements: Directed Weapons



- **High attack precision:** lower collateral damage and civilian casualties.
- **Reduction of human risk** by being operated remotely.
- **Extended operations** without having to worry about fatigue.
- **Versatility:** equipped with a variety of weapons, from missiles to bombs.

Drones as War Elements: Symmetrical Warfare



- **Lower cost effectiveness** compared to traditional manned aircraft.
- **Cost reduction in drone production** (cardboard drones, wooden drones, 3D printing, cheap chipsets, etc.)

Drones as War Elements: Strategy



- **Means of strategy and tactics** for real-time covert operations.
- **Surveillance capability and data collection** for intelligence purposes.

Use of drones for SIGINT (Signals Intelligence)



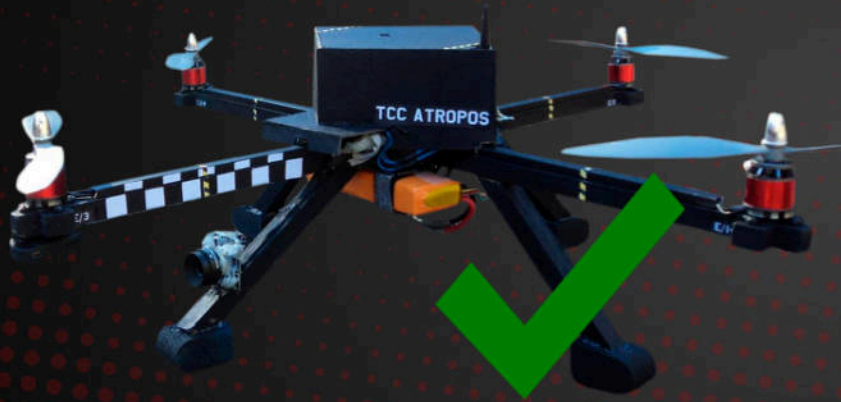
- **Collection of information**, interception, and analysis of enemy communications and electronic signals.
- **Extended reach** to hard-to-access areas.
- **Prolonged operations** continuously collecting data.
- **Real-time transmission of information** for analysis, enabling quick response to emerging threats.
- **Covert operations** from high altitudes or in discreet flight patterns.

Anti-drone measures



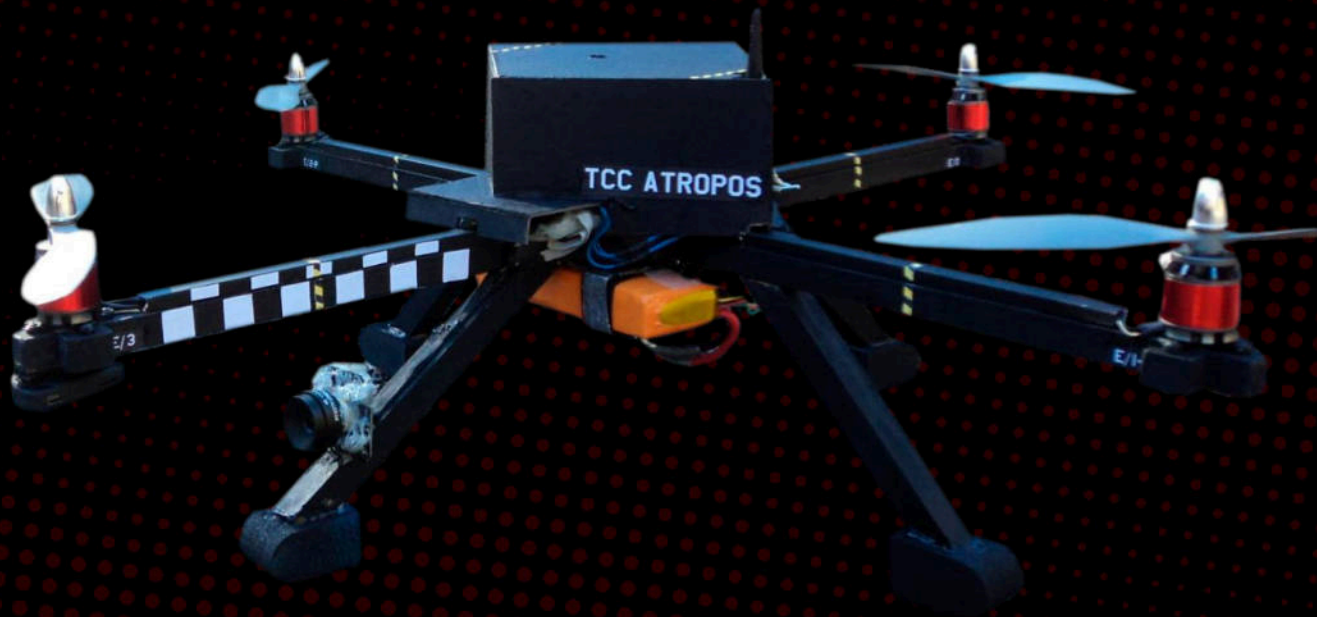
- **Thermal and standard cameras** for shape detection.
- **Detection of heat** emitted by engines and electronics.
- **Characterization** of drone noise.
- **Radio frequency and waveform** detection.
- **Radio signature.**

Counter-Countermeasures



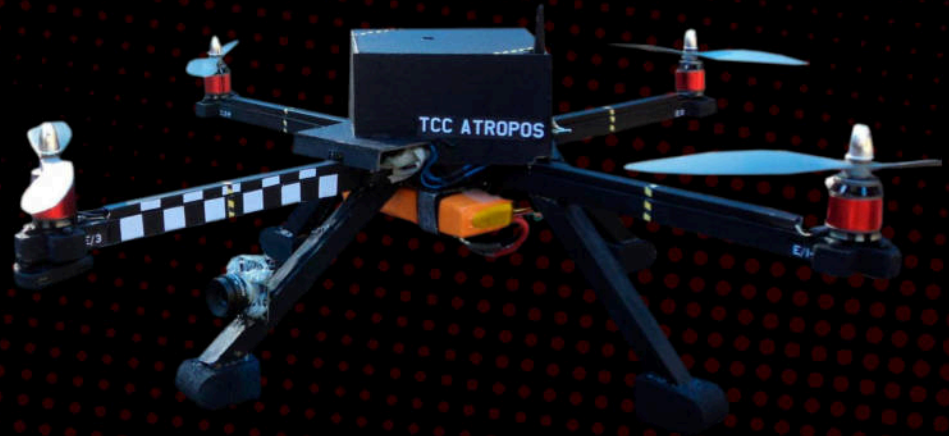
- Frequency hopping
- Utilization of **unexpected frequencies** by the jammer
- Robust protocols
- Physical attacks

EVOLUTION OF 'ATROPOS' AS A WAR WEAPON



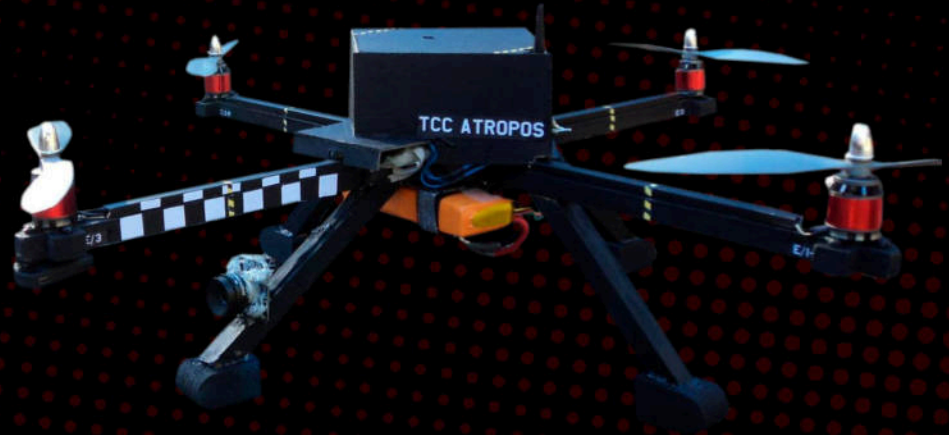
ATROPOS

- WiFi Router
- PIC16F876 for PWM
- Wii Nunchuck and
- Motion + as onboard IMU
- HTML5 telemetry by router
- Webserver
- WiFi
- WPS Attacks with bully



EVOLUTION OF ATROPOS: STAGE 1

- Avoiding deauthentication attacks with **Beacon Frames**
- Using beacon frames to **deceive adversary SIGINT** by creating fictitious drones

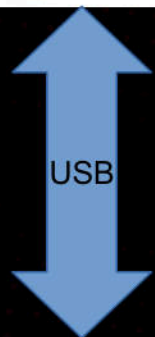


EVOLUTION OF ATROPOS: STAGE 2

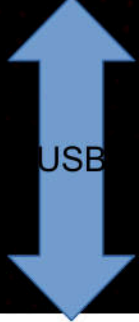
- Testing a **new concept of mini-drone (INTERCEPTOR)** with the ATROPOS model.
- Adding autonomous hacking capabilities **using the drone's own hardware.**
- Incorporating a **control side-channel using an FM transmitter.**



2012-2014



2014-2017



USB



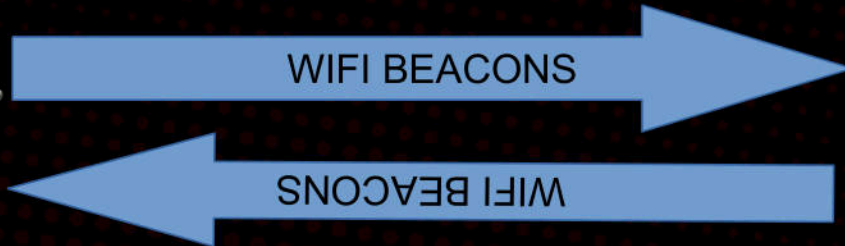
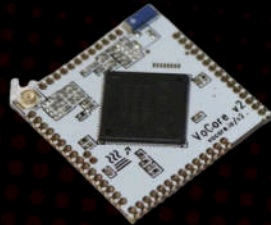
WIFI BEACONS



2018



USB



WIFI BEACONS

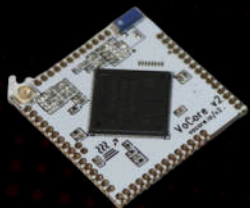
WIFI BEACONS



2020



USB



LAN



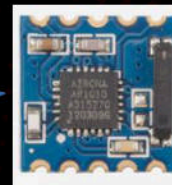
WIFI BEACONS

WIFI BEACONS

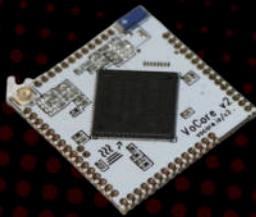
FM RDS PAYLOAD



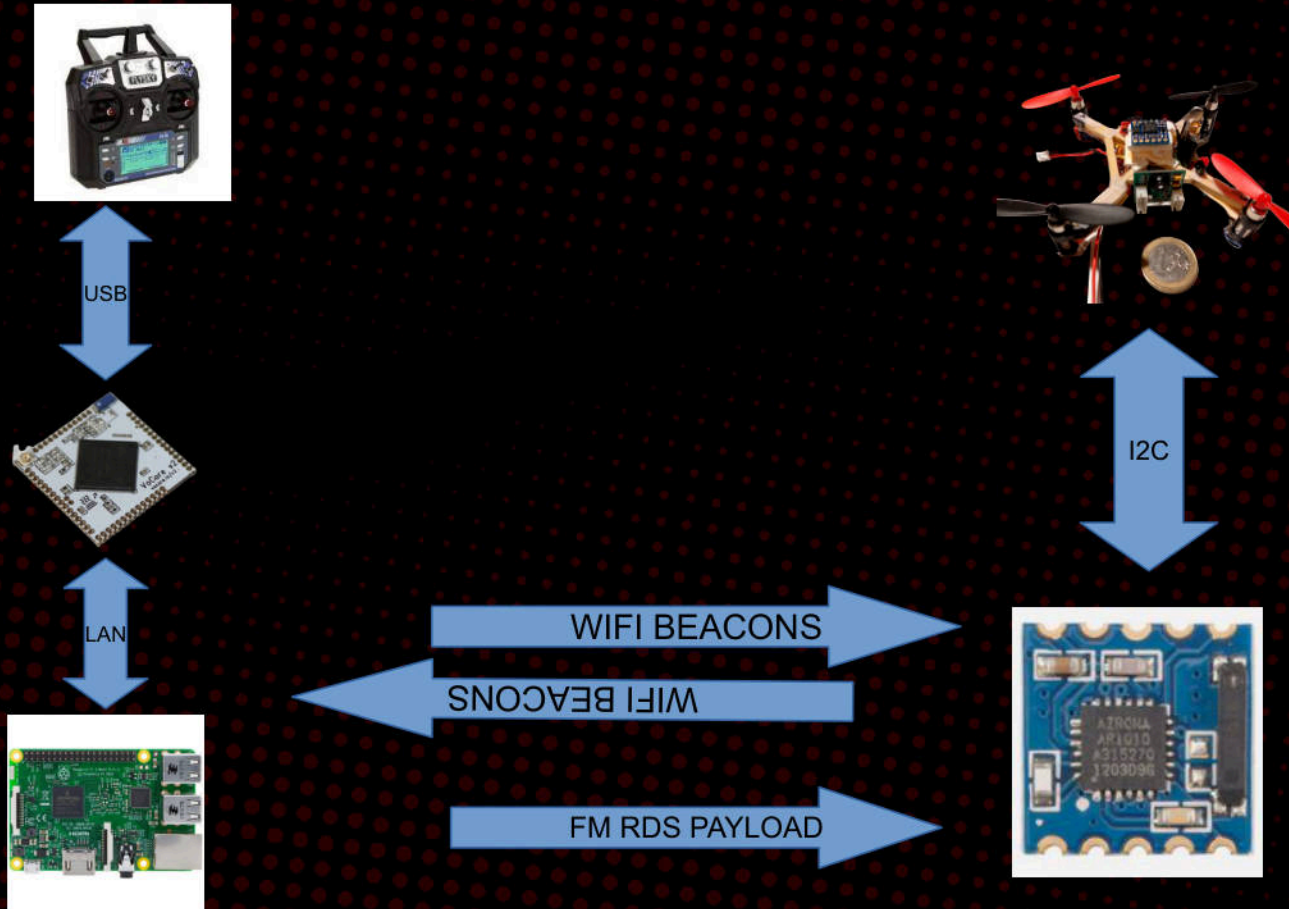
I2C



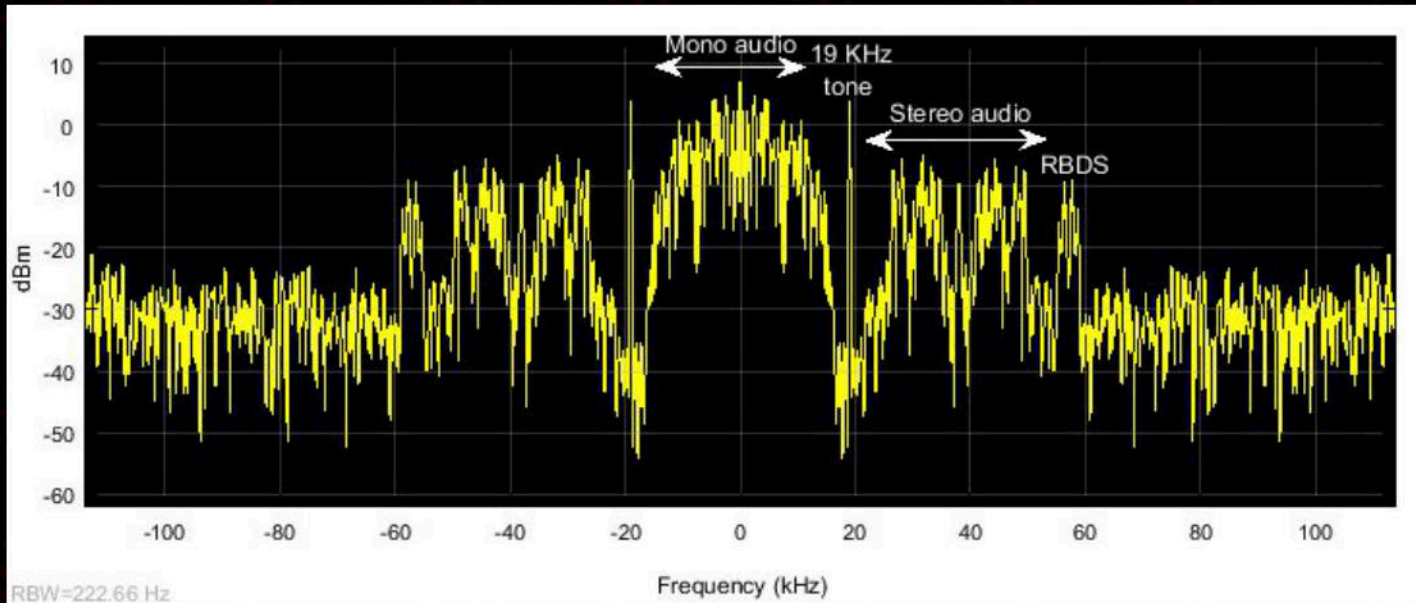
Associated Experiments: Interceptor



Associated Experiments: Interceptor



RDS FM



EVOLUTION OF ATROPOS: STAGE 3

WAR WEAPON



EVOLUCIÓN DEL ATROPOS: STAGE 3

➤ **Step 1:** Controlling the drone through arbitrary frequency generation



EVOLUCIÓN DEL ATROPOS: STAGE 3

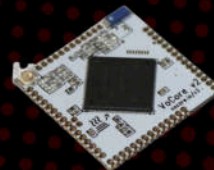
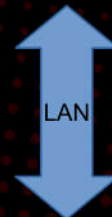
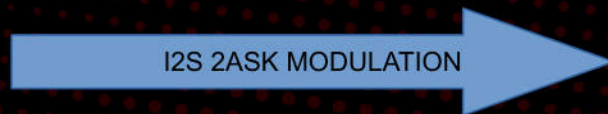
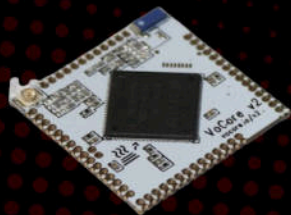
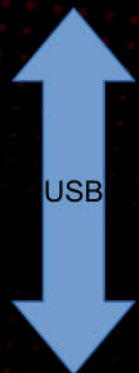
➤ **Step 2:** Signal reception using SDR (Software-Defined Radio) within the drone.



EVOLUCIÓN DEL ATROPOS: STAGE 3

➤ **Step 3:** Application of spread spectrum using frequency hopping method.





SPREAD SPECTRUM

Spread spectrum is a **modulation technique that distributes the signal over a much wider frequency band** than what is required by the original information.

The use of spread spectrum **improves resistance to noise and interference**, as a signal spread over a wider spectrum can better tolerate noise at a specific frequency.

SPREAD SPECTRUM

- FHSS

Frequency Hopping Spread Spectrum

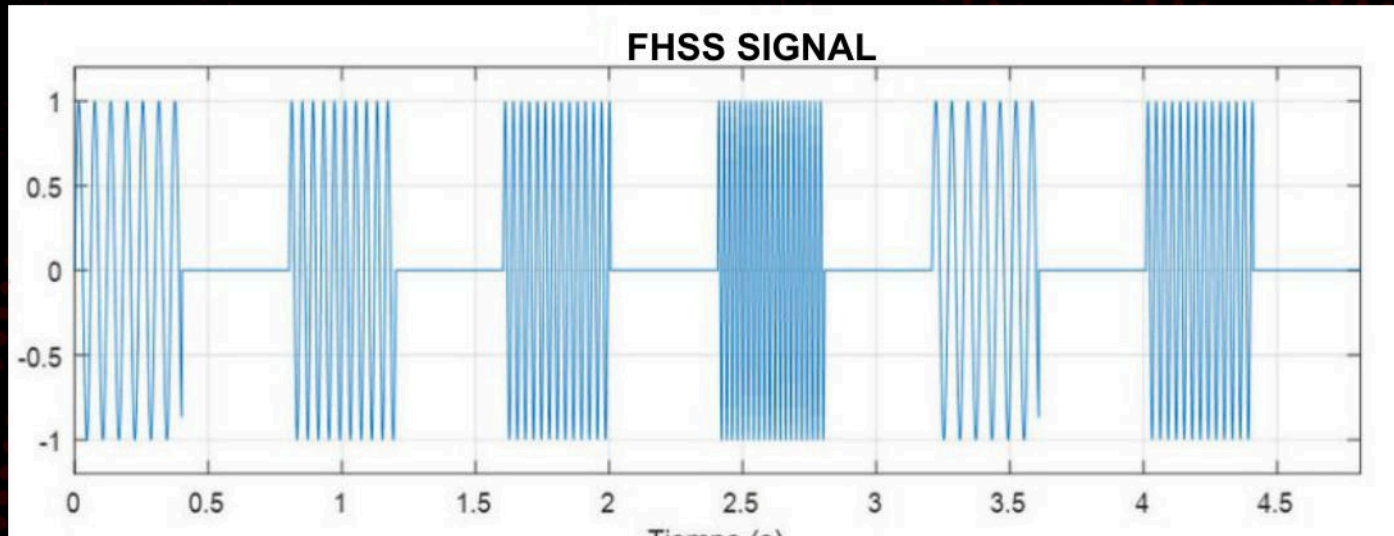
- DSSS

Direct Sequence Spread Spectrum



FREQUENCY HOPPING SPREAD SPECTRUM (FHSS)

In FHSS, the data signal is transmitted over different frequencies in a predetermined pattern that only the receiver knows and can track.



**¿Would it be possible with an
MT7628?**

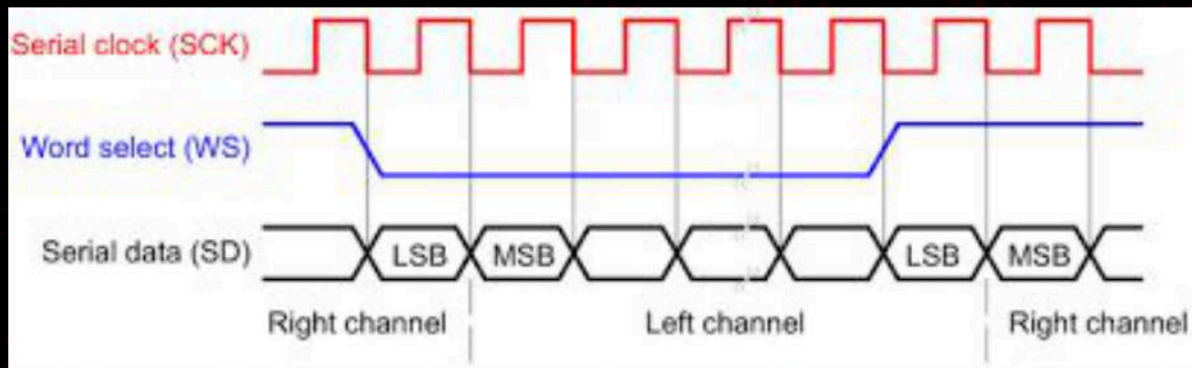
HOLD MY



BEER

I2S BUS

- Audio digital interface
- ***IT IS NOT I2C BUS***
- CLOCK PIN
- PIN DATA TX
- PIN DATA RX



I2S Schema in Mediatek MT7628

I2S

16	I2S_SDI	O	4 mA	I2S data input
17	I2S_SDO	I/O, IPD	4 mA	I2S data output
18	I2S_WS	O	4 mA	I2S word select
19	I2S_CLK	I/O	4 mA	I2S clock

1000.0A00

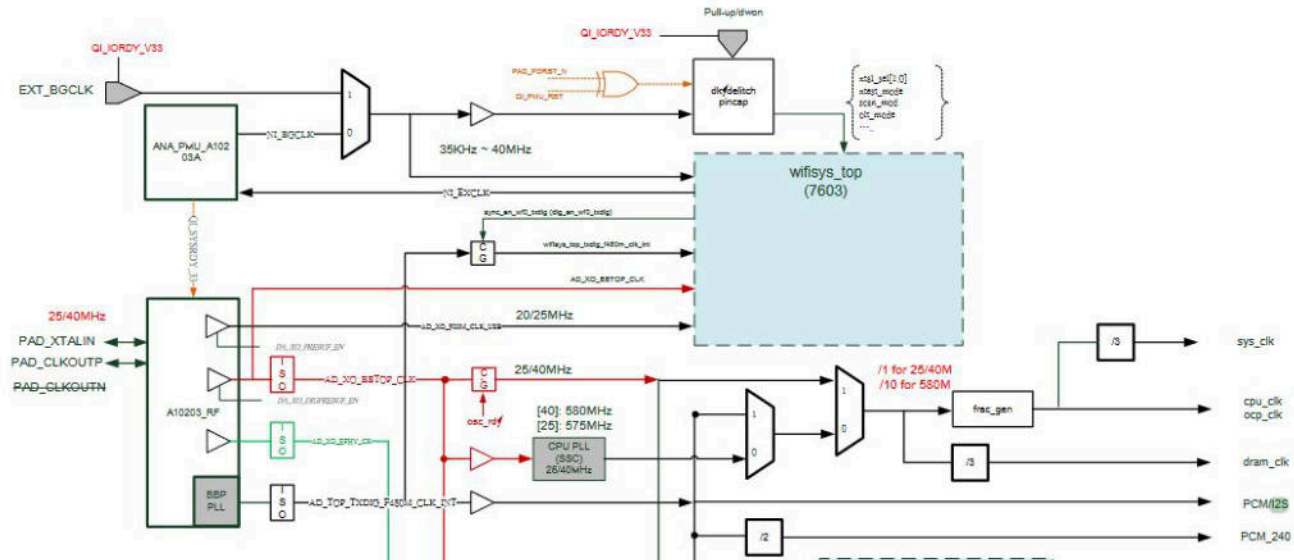
-

1000.0AFF

256 Bytes

I2S

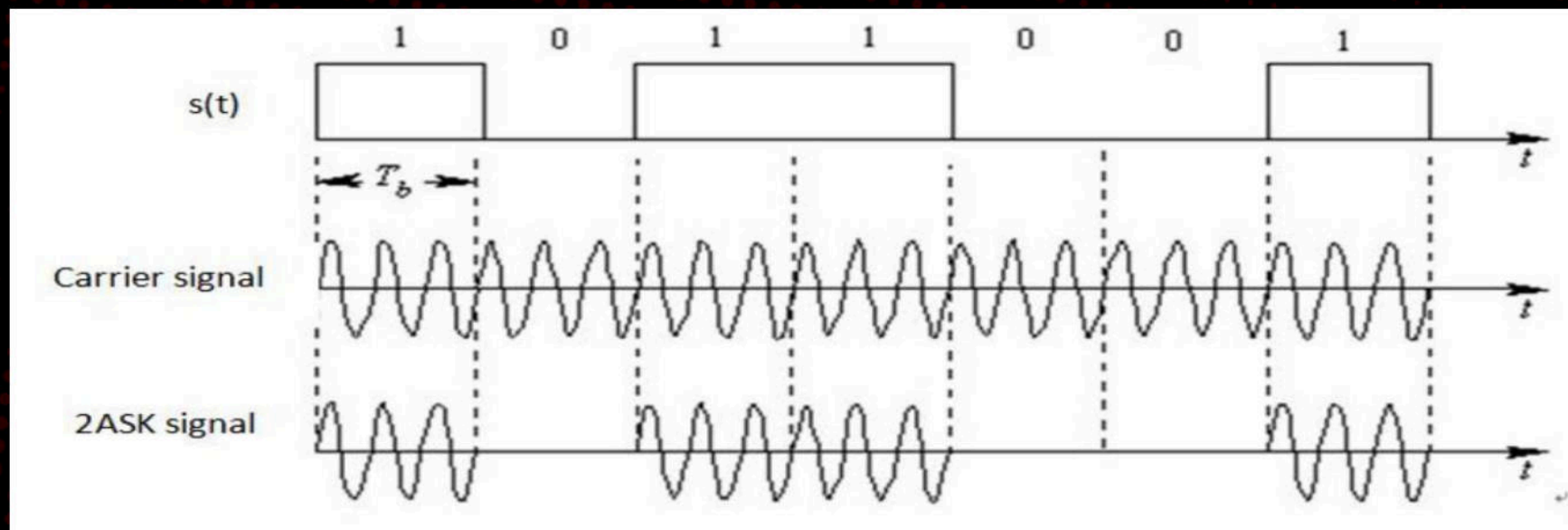
1.4 Clock Plan



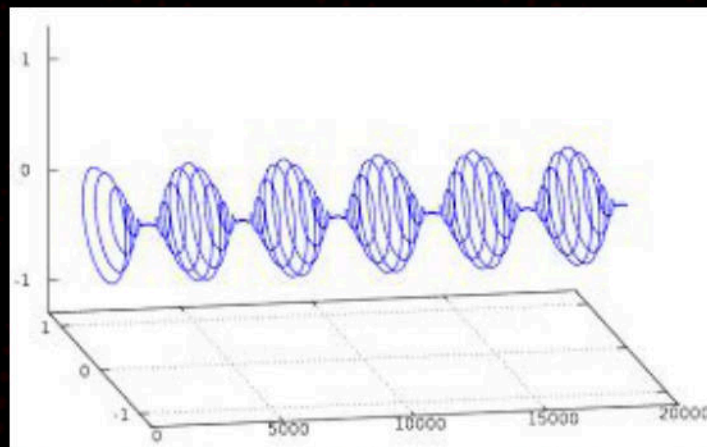
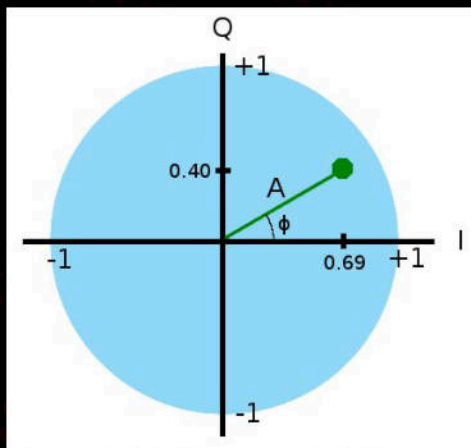
Module name: I2S Base address: (+10000A00h)

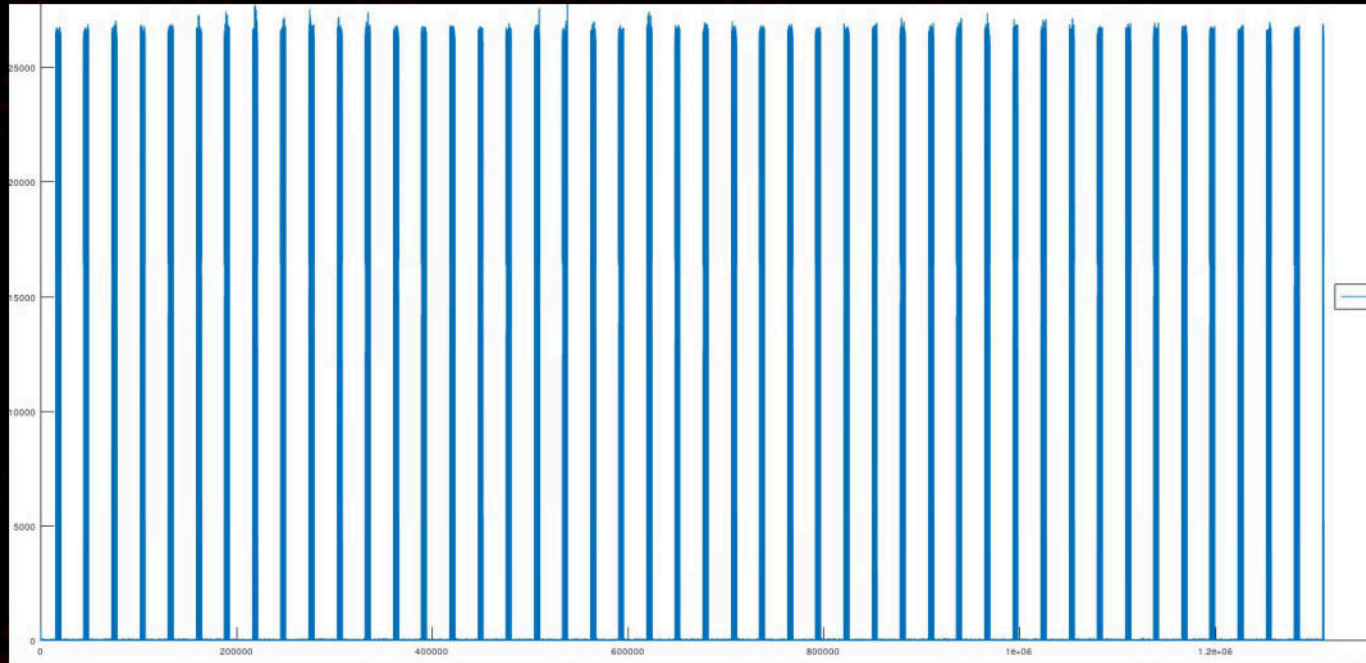
Address	Name	Width	Register Function
10000A00	<u>I2S_CFG</u>	32	I2S Configuration I2S Tx/Rx Configuration Register
10000A04	<u>INT_STATUS</u>	32	Interrupt Status I2S Interrupt Status
10000A08	<u>INT_EN</u>	32	Interrupt Enable I2S Interrupt Enable Control Register
10000A0C	<u>FF_STATUS</u>	32	FIFO Status I2S Tx/Rx FIFO Status
10000A10	<u>TX_FIFO_WREG</u>	32	Transmit FIFO Write to Register Tx Write Data Buffer
10000A14	<u>RX_FIFO_RREG</u>	32	Receive FIFO Read Register DRAM PAD CONTROL 3
10000A18	<u>I2S_CFG1</u>	32	I2S Configuration 1 I2S Loopback Test Control Register
10000A20	<u>DIVCOMP_CFG</u>	32	Integer Part of the Divisor Register 1 Integer Part of the Divisor Register
10000A28	<u>DIVINT_CFG</u>	32	Integer Part of the Divisor Register 2 Integer Part of the Divisor Register

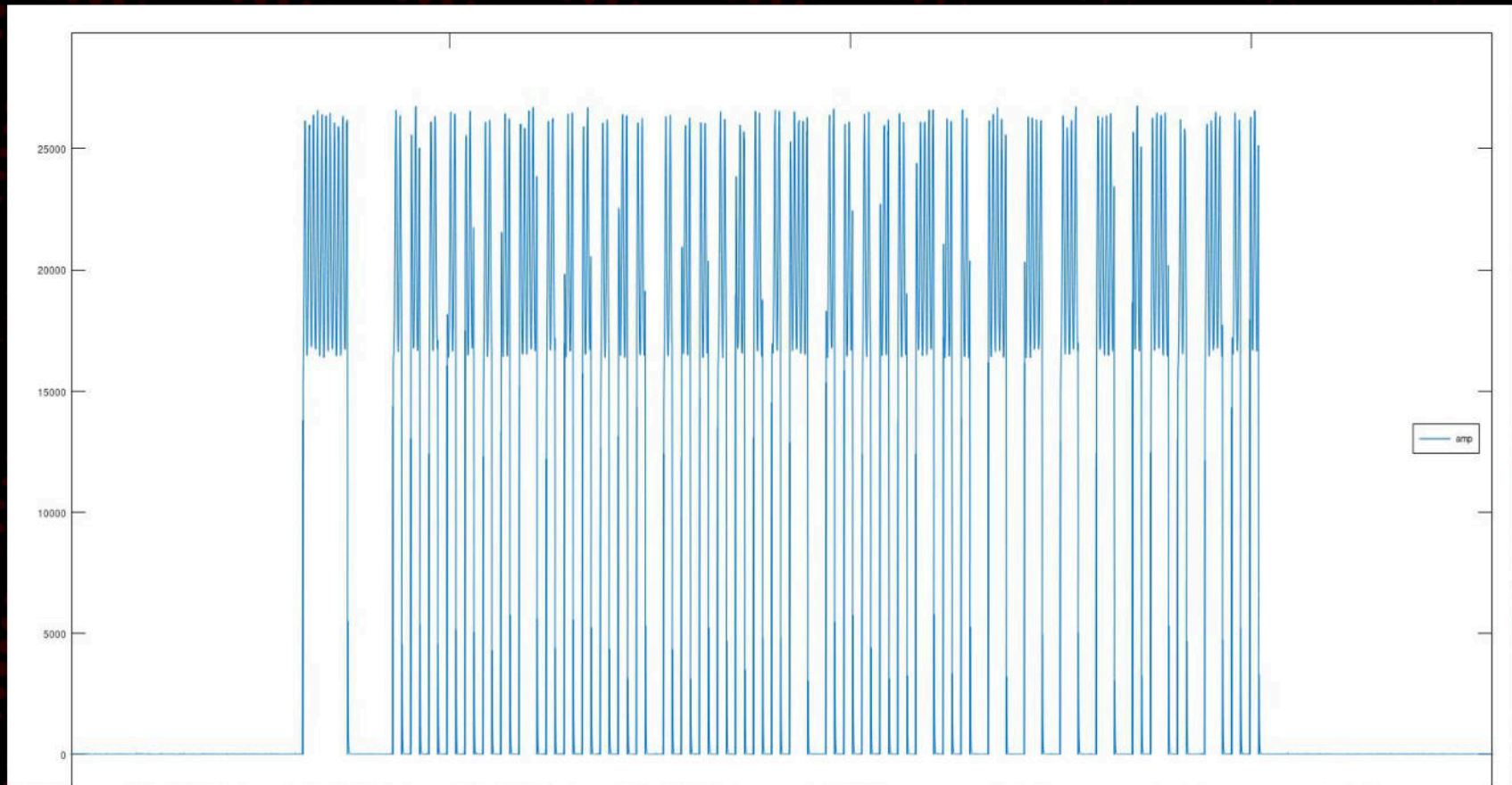
2ASK



In-phase / Quadrature

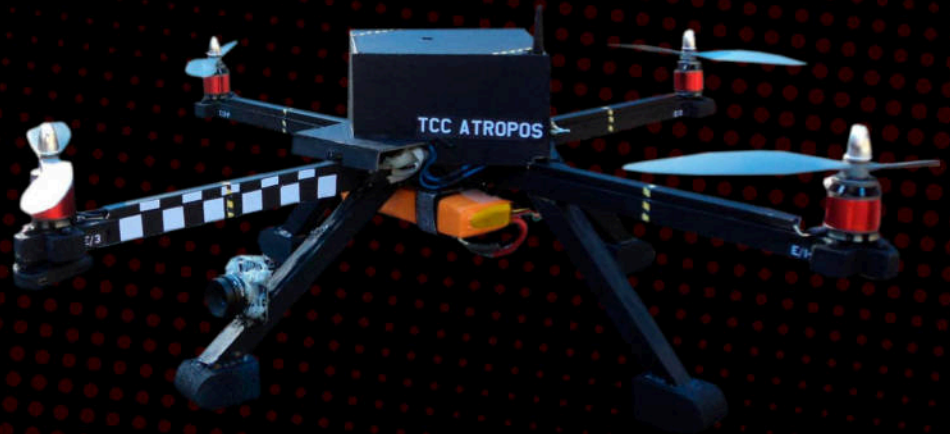




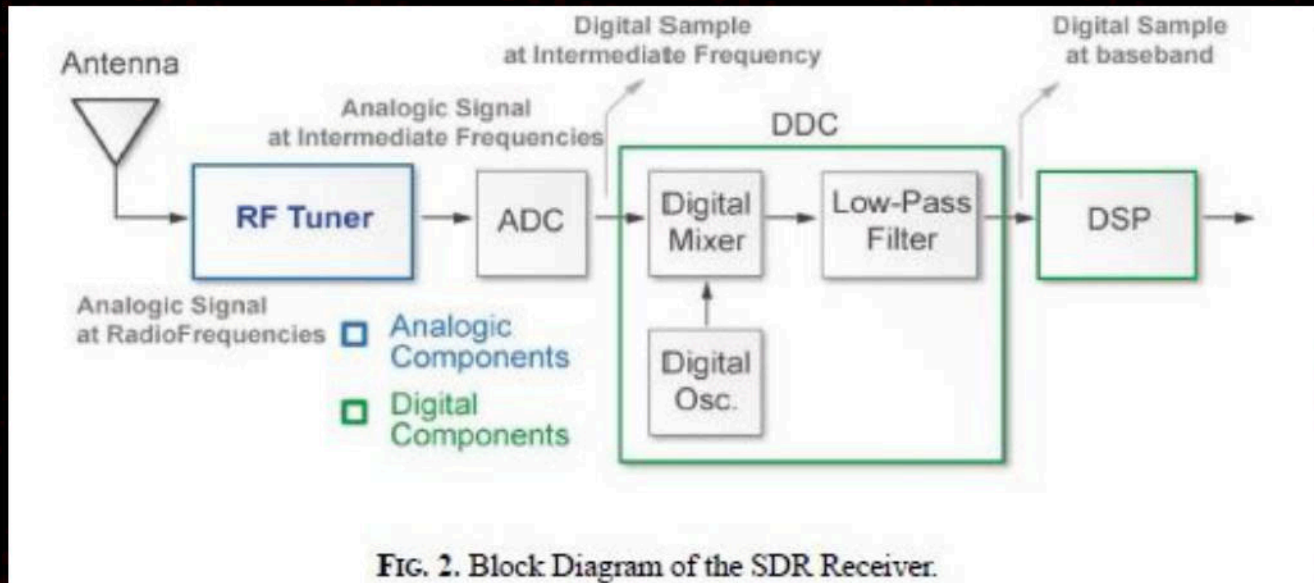


Custom Driver for I2S de MT7628

- Accepts bit stream from a dev file (/dev/fh_ss)
- Modulates in "2ASK". Adds CRC8
- TWO hrtimers. One for symbol generation and the other one for packet timed synced generation



RECEIVER: Another Vocore2, with SDR





HOW I DID IT

By

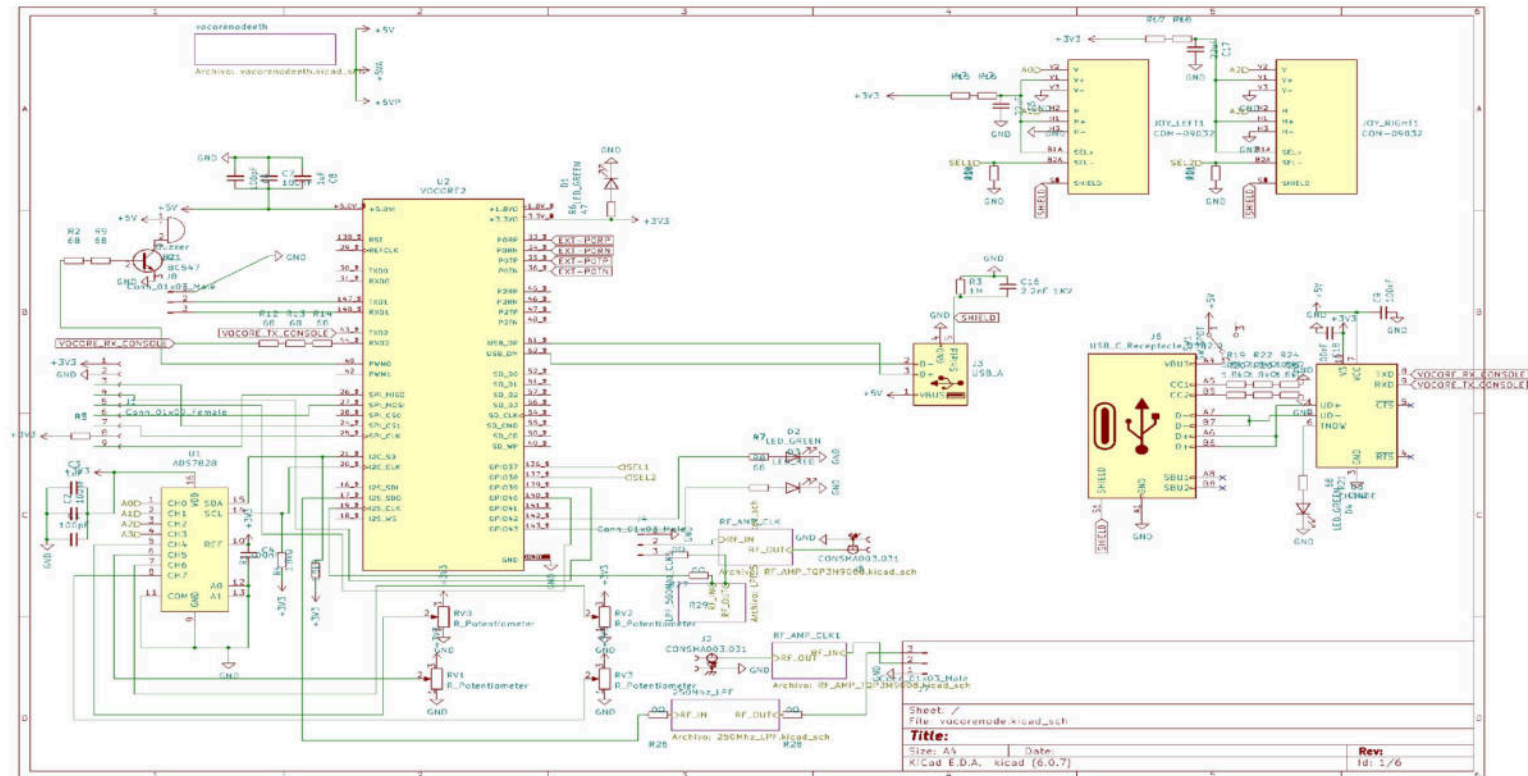
Victor Frankenstein

David Meléndez

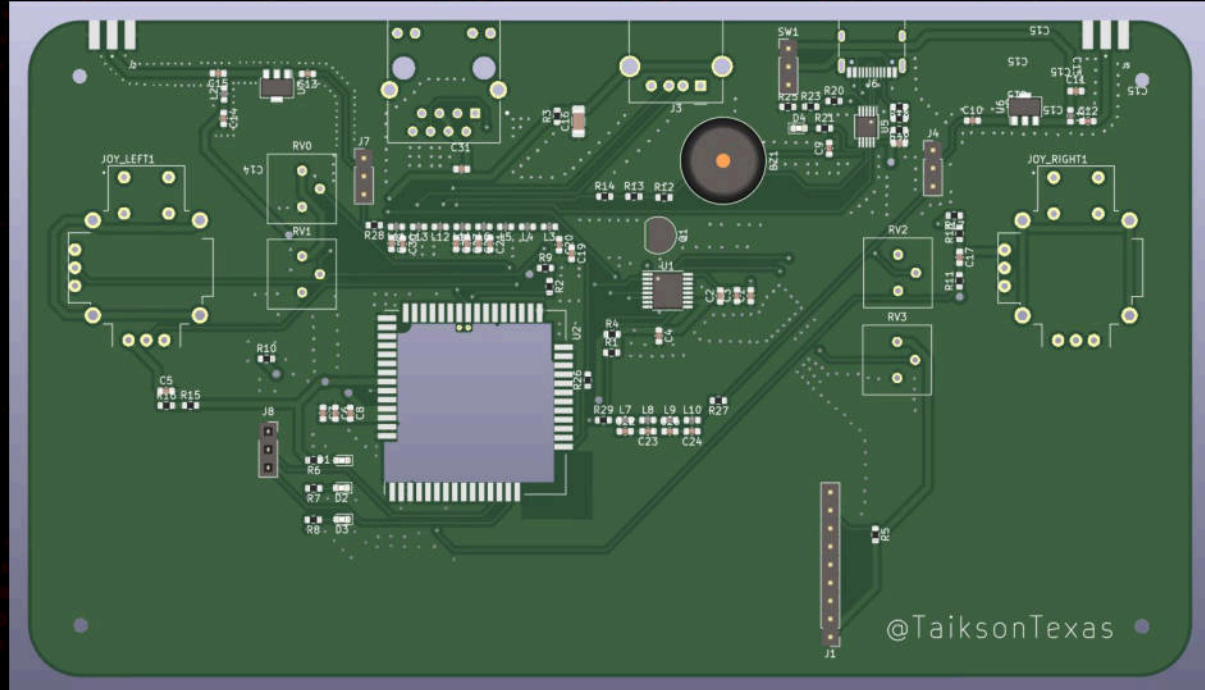
SDR: Frequency hopping

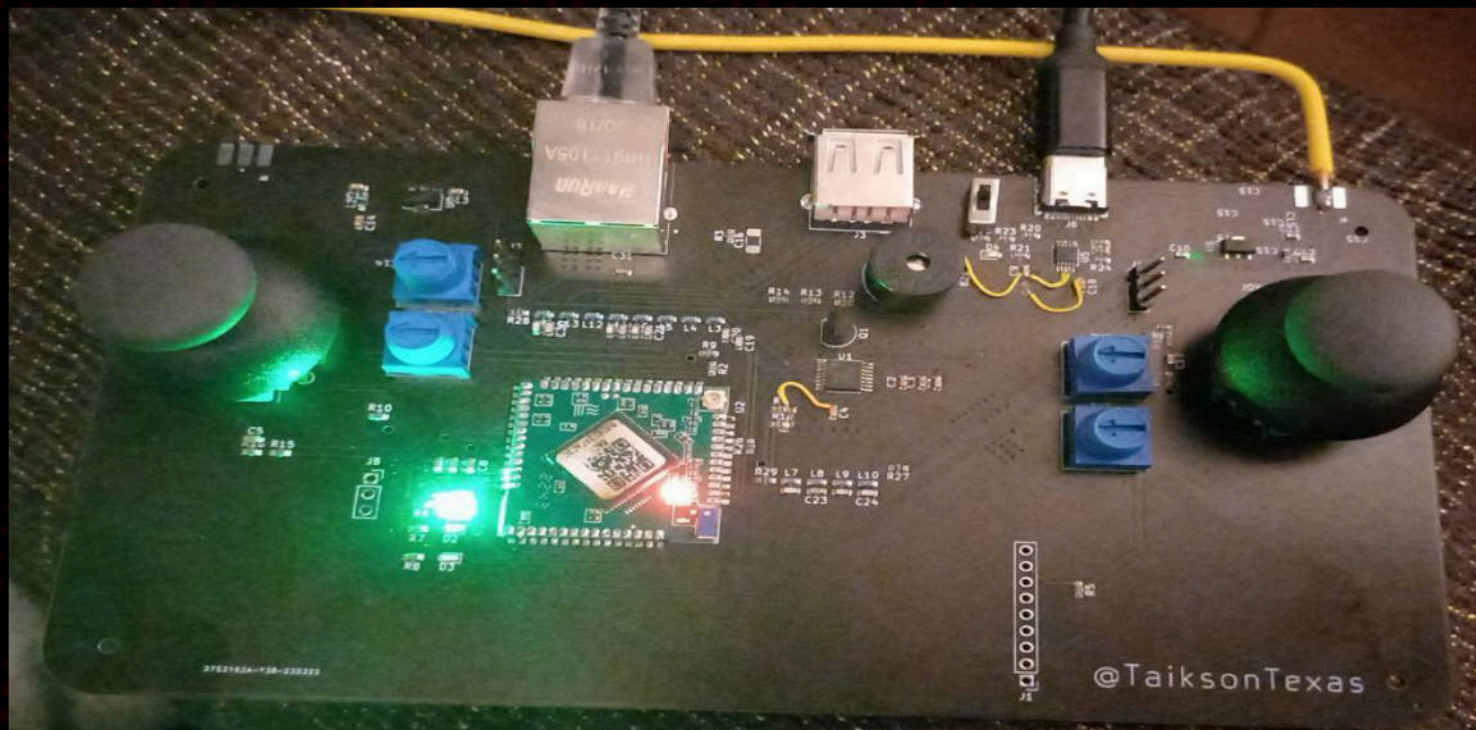
- Requires reiterative calls to librtlsdr
`rtlsdr_set_center_freq(dev, divisor_to_freq_hz(_divisors_fhss[0]));`
- Synchronization with pregenerated and preshared list of frequencies.

Custom controller with all components:



Custom controller with all components:





Next episode: DSSS

Thanks!



@TAIKSONTEXAS



@CONSTRAINTERROR